# Mohamed Amine Bayar, PhD | Independent Biostatistics Consultant | Oncology & Hematology

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English, French & Arabic

## Profile

- PhD-level biostatistics consultant with 13+ years across the full clinical development lifecycle — from Bayesian Phase I dose-escalation through Phase III pivotal design, high-stakes analysis milestones, and regulatory submission.
- A profile combining frontline industry experience (Johnson & Johnson, Novartis), academic research depth (PhD Paris-Saclay, 15+ publications, 1,200+ citations), hands-on ML/AI expertise applied to clinical and genomic data (co-invented patent WO2018002385A1), and advanced programming capabilities (R/Shiny, Python, SAS). Direct FDA and EMA interactions on pivotal trial design and regulatory strategy.
- Core expertise in oncology and hematology (multiple myeloma, CML, solid tumors — 8+ disease areas); methodology and regulatory experience transferable across therapeutic areas.

## Technical Expertise

- **Statistical Methodology:** Survival analysis (Cox, competing risks, frailty models); linear, logistic, and mixed-effects models; Bayesian dose-escalation (CRM, BLRM, BOIN); group sequential and adaptive designs (seamless Phase II/III, platform trials); multiple testing procedures; Estimand framework (ICH E9(R1)); sample size calculation and event re-estimation; penalized regression (Lasso, Ridge, Elastic Net); PKPD and exposure-response modeling.
- **Programming — R:** tidyverse, ggplot2, RShiny, RMarkdown; OncoBayes2 (Bayesian BLRM dose-escalation); rpact, gsDesign (group sequential designs); survival, cmprsk (time-to-event); automated TFL pipeline development.
- **Programming — Python:** pandas, NumPy, scikit-learn (ML/AI); lifelines, scikit-survival (survival analysis); XGBoost, SHAP (predictive modeling and interpretability).
- **Programming — Other:** SAS (Base, Stat, Macro), SQL; nlmixr (PKPD/tumor growth modeling, R).
- **ML/AI:** Predictive modeling and prognostic signature development; variable selection in high-dimensional genomic spaces; internal and external model validation (cross-validation, bootstrap); co-invented patent applying ML to oncology outcomes (WO2018002385A1).
- **Medical Writing:** SAPs, TFL shells, CSRs, SCS/SCE, briefing documents, health authority responses, IBs.

## Work Experience

**BioSpark** — **Remote, France**
*Independent Biostatistics Consultant* — *Apr 2026 – Present*
Provides independent statistical consulting across the full drug development lifecycle, from early-phase design through regulatory submission. Developing expertise at the intersection of classical biostatistics and AI/ML methodologies.

**Johnson & Johnson** — **Remote, France**
*Statistician — Oncology & Hematology, 3 years & 2 months* — *Feb 2023 – Mar 2026*

Trial statistician across four oncology programs (multiple myeloma and NSCLC), spanning statistical design, 10+ high-stakes analysis milestones, and regulatory submission.

- **Analysis delivery (teclistamab, ciltacabtagene autoleucel):** Led statistical delivery of multiple analysis packages — interim analyses, safety snapshots, and primary analyses — across two cutting-edge myeloma programs in Phase II multi-arm settings: teclistamab (BCMA×CD3 bispecific); and ciltacabtagene autoleucel (BCMA-directed CAR-T). Provided CRO statistical oversight on two studies and contributed to onboarding and mentoring of new statisticians joining the team.
- **Statistical design and study conduct (ramantamig):** Served as lead statistician for Phase Ib and Phase III design on a first-in-class BCMA×GPRC5D×CD3 trispecific — including BOIN dose-escalation design and simulation, Phase III event re-estimation, interim efficacy analyses, OS harm monitoring, and direct FDA/EMA interactions from concept through regulatory alignment (EOP2 briefing book submission and governance activities).
- **Regulatory submission (amivantamab):** Contributed to the submission package for amivantamab in NSCLC — extending regulatory experience beyond myeloma.
- Authored Protocols statistical sections, SAPs, TFL shells, CSRs, and regulatory briefing documents; prepared written responses to FDA/EMA health authority queries.

**Novartis** — **Paris, France**

*Statistician — CML Portfolio (Scemblix), 3 years 3 months* — *Nov 2019 – Jan 2023*

- **Phase II statistical delivery (asciminib):** Led full statistical delivery for a 4-arm Phase II study — from SAP development through database lock, primary analysis, and reporting — forming part of the integrated evidence base supporting the approved label.
- **Pediatric regulatory program (PIP/iPSP):** Led statistical methodology for a Phase Ib adult-to-pediatric bridging study developed under the joint EMA/FDA pediatric framework (Paediatric Investigation Plan and initial Pediatric Study Plan) — a mandatory regulatory requirement for novel oncology agents. Implemented Bayesian BLRM dose-escalation using the OncoBayes2 R package, including prior calibration from adult data and simulation to support dose recommendation in a pediatric population.
- **Pharmacometrics rotation (formal secondment):** Completed a 7-month full-time rotation within the Novartis pharmacometrics team — nlmixr R package.

**Gustave Roussy Cancer Center** — **Paris, France**

*Statistician — Solid Tumors, 6 years 9 months* — *Mar 2013 – Oct 2019*

Trial statistician on 7 clinical trials (Phase I to III) and multiple translational research projects, with responsibility for trial delivery, methodological research, and scientific publication.

- **Trial delivery and analysis milestones:** Led statistical activities on named programs including POP, CHIPASTIN, HPVRX, MELIPI RX, PACS04, IGRT-P, and others — spanning breast (TNBC, HR+, metastatic, early), ovarian, melanoma, prostate, and pediatric oncology. Delivered full analysis lifecycles including interim, primary, and final analyses.
- **Translational research & prognostic modeling:** Developed and validated multiple prognostic models across tumor types — including three sarcopenia prognostic models, a risk stratification model in pediatric intracranial ependymoma, and a survival prognostic model in second-line metastatic renal cell carcinoma.
- **Precision medicine & platform trials (SAFIR02, MAPPYACTS):** Substantial contributor to two landmark precision medicine genomic screening platforms — delivering fully automated TFL pipelines for IDMC meetings. PhD research (Part 3) focused on optimal statistical analysis strategies for platform trials such as SAFIR02.
- **Co-invented patented ML pipeline:** Built and validated a genomic prognostic signature for TNBC (Patent WO2018002385A1) — owning the full pipeline from raw gene expression extraction through data normalization (fRMA), cross-platform merging, predictive modeling, and internal/external model validation; published in *Annals of Oncology.*
- **Methodological research & academic output:** Generated 15+ peer-reviewed publications (1,200+ citations) across *Annals of Oncology*, *JCO*, *CCR*, *Statistics in Medicine*; Invited visiting scientist at Mayo Clinic, hosted by Prof. Daniel J. Sargent. Taught biostatistics and public health to undergraduates at Paris Sud University.

## Education

**Paris Saclay University** — **Paris, France**

*PhD in Public Health — Major Biostatistics* — *Oct 2016 – Nov 2019*

- Topic: Long-term evaluation of randomized Phase III clinical trials in oncology with rare diseases and biomarker-based subtypes.
- Developed novel methodology for randomized trials with seamless Phase II/III, platform trials, and interim analysis strategies; and produced 3 first-author publications in *Statistics in Medicine*, *Statistical Methods in Medical Research*, and *Computer Methods and Programs in Biomedicine.*

**ESSAI — Université de Carthage** — **Tunis, Tunisia**

*Engineer in Statistics* — *Sep 2010 – Jun 2013*

Diplôme National d'Ingénieur — Statistique et Analyse de l'Information

**ENIT — Université El Manar** **Tunis, Tunisia**
*MSc in Computational Biology* *Sep 2012 – Oct 2013*
Diplôme National de Masterè de Recherche — Traitement de l'Info. et Complexité du Vivant

## Intellectual Property & Innovation

**Patent**: WO2018002385A1 — hands-on ML application to medicine: built and validated a genomic signature to predict prognosis in Triple Negative Breast Cancer, owning the full pipeline from raw gene expression data to a clinically validated predictive model. https://patents.google.com/patent/WO2018002385A1/en

## Selected Publications

**2020**: **Mohamed Amine Bayar**, Gwénaël Le Teuff, Franz Koenig, Marie-Cécile Le Deley, Stefan Michiels. *Group sequential adaptive designs in series of time-to-event randomized trials in rare disease: a simulation study.* Statistical Methods in Medical Research, Jun. 2020. https://pubmed.ncbi.nlm.nih.gov/31354106

**2016**: **Mohamed Amine Bayar**, Gwénaël Le Teuff, Stefan Michiels, Daniel J Sargent, Marie-Cécile Le Deley. *New insights into the evaluation of randomized controlled trials for rare diseases over a long-term research horizon: a simulation study.* Statistics in Medicine, Aug. 2016. https://www.ncbi.nlm.nih.gov/pubmed/27027783

**2018**: C Criscitiello, **MA Bayar**, G Curigliano, FW Symmans, C Desmedt, H Bonnefoi, B Sinn, G Pruneri, Vicier, Pierga, Denkert, S Loibl, C Sotiriou, S Michiels, F André. *A gene signature to predict high tumor-infiltrating lymphocytes after neoadjuvant chemotherapy and outcome in patients with triple-negative breast cancer.* Annals of Oncology, Jan. 2018. https://www.ncbi.nlm.nih.gov/pubmed/29077781

**Full list**: 15+ papers, 1,200+ citations: https://scholar.google.com/citations?user=u6Ns41gAAAAJ&hl=en